

Richie Tran

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EDUCATION

Carnegie Mellon University

B.S. Computer Science and Machine Learning

Pittsburgh, PA

May 2027

Relevant Coursework

Data Structures and Algorithms, Computer Systems, AI Representation and Problem Solving, Machine Learning I, Discrete Mathematics, Functional Programming, Ideas in Theoretical Computer Science

EXPERIENCE

Software Engineer Intern | *Opportunity Knocks*

December 2025 - Present

- Developed a React Native mobile app with Supabase to bring the existing web application to iOS and Android, targeting 5,000+ users at launch.
- Improved application stability by resolving production bugs, reducing reports, and shipped new features to enhance user experience.

Research Assistant | *Reliable Autonomous Systems Lab, CMU*

January 2025 - Present

- Developing Victor the GameBot in Python, a Scrabble-playing agent integrating computer vision, speech recognition, and audio processing for real-time human-robot gameplay and interaction.
- Building a computer vision pipeline to detect and interpret board state, and implementing speech and audio processing modules for natural language trash-talk during gameplay.

Software Developer | *Scotty Labs, CMU*

August 2024 – Present

- Built an AI-powered presentation tool in React and Node.js, enabling 100+ users to receive real-time feedback on delivery.
- Improved campus accessibility for 1,000+ students by contributing to CMUMaps, CMUResearch, and CMUEats.

Driverless Control Engineer | *Carnegie Mellon Racing*

August 2024 – Present

- Reduced path-planning latency by optimizing algorithms in C, integrating sensor fusion for autonomous navigation across multiple track configurations.
- Contributed to team's 13th place finish out of 100 teams at Formula FSAE Elecric.

PROJECTS

MyEyes | *Python, Swift, Flask, CoreML, YOLOv8*

- Developed an audio-based iOS app enabling visually impaired users to navigate safely through real-time hazard detection.
- Achieved 90% detection accuracy by deploying a fine-tuned YOLOv8 model via CoreML with a custom urgency-weighted loss function.
- Integrated LiDAR and camera-based obstacle identification with Siri commands for accessibility.

StoryScroll | *Python, JavaScript, React, FastAPI, FFmpeg, ElevenLabs API, Cloudflare R2*

- Engineered a full-stack video generation pipeline converting Reddit stories into short-form social media content by compositing ElevenLabs TTS narration with background footage via FFmpeg.
- Designed an asynchronous job queue with real-time progress polling, enabling concurrent video processing across a React frontend and FastAPI backend.
- Implemented story sourcing using Reddit scraping, with voice selection, acronym normalization, and Cloudflare R2 cloud storage for scalable background video delivery.

AI Logo Generator | *Node.js, Express, Claude API, JavaScript, HTML, CSS*

- Built a full-stack web app that processes user meeting transcripts and generates custom SVG logos and design rationales, leveraging Claude AI to streamline brand identity creation.
- Developed RESTful API endpoints with Express.js to integrate Claude Sonnet 4.5 and Opus 4.5 models for real-time brand analysis and logo generation.

SKILLS

Programming Languages: C, C++, Python, Swift, Java, JavaScript, HTML, CSS, TypeScript, Bash

Tools: Node.js, Express, React, Claude API, Flask, FastAPI, Firebase, Supabase, React Native, Git